



# Propositional Logic and Predicate Logic

# Join the class

## Proposition

Statement which is either true or false  
but not both at the same time.

## Propositional variable

Variable indicating proposition  $\Rightarrow$  Propositional variable  
will hold a truth  
value — T or F.

| p |
|---|
| T |
| F |

}  $2 = \underline{\underline{2^1}}$

| p | q |
|---|---|
| T | T |
| T | F |
| F | T |
| F | F |

}  $4 = \underline{\underline{2^2}}$

| p | q | r |
|---|---|---|
| T | T | T |
| T | T | F |
| T | F | T |
| T | F | F |
| F | T | T |
| F | T | F |
| F | F | T |
| F | F | F |

}  $8 = \underline{\underline{2^3}}$

'n' propositional variables



$2^n$  truth combinations

## Logical operators (Connectives)

- AND (Conjunction) ( $\wedge$ )
- OR (Disjunction) ( $\vee$ )
- NOT (Negation) ( $\sim$ ) ( $\neg$ )
- XOR (Exclusive OR) ( $\oplus$ )
- Implication ( $\rightarrow$ )
- Biimplication ( $\leftrightarrow$ )
- $\vdots$

- NAND ( $\uparrow$ )
- NOR ( $\downarrow$ )
- $\vdots$

| $p$ | $q$ | $\neg p$ | $\neg q$ | $p \wedge q$ | $p \vee q$ | $p \rightarrow q$ | $p \oplus q$ | $p \uparrow q$ | $p \downarrow q$ |
|-----|-----|----------|----------|--------------|------------|-------------------|--------------|----------------|------------------|
| T   | T   | F        | F        | T            | T          | T                 | F            | F              | F                |
| T   | F   | F        | T        | F            | T          | F                 | T            | T              | F                |
| F   | T   | T        | F        | F            | T          | T                 | T            | T              | F                |
| F   | F   | T        | T        | F            | F          | T                 | F            | T              | T                |

$$p \oplus q \equiv (p \wedge \neg q) \vee (\neg p \wedge q)$$

| $p$ | $q$ | $p \oplus q$ | $\neg p$ | $\neg q$ | $(p \wedge \neg q)$ | $(\neg p \wedge q)$ | $(p \wedge \neg q) \vee (\neg p \wedge q)$ |
|-----|-----|--------------|----------|----------|---------------------|---------------------|--------------------------------------------|
| T   | T   | F            | F        | F        | F                   | F                   | F                                          |
| T   | F   | T            | F        | T        | T                   | F                   | T                                          |
| F   | T   | T            | T        | F        | F                   | T                   | T                                          |
| F   | F   | F            | T        | T        | F                   | F                   | F                                          |

Create formula for Exclusive OR using these truth combinations.

Same

$$P \leftrightarrow q$$

| P | q | $P \rightarrow q$ | $q \rightarrow P$ | $(P \rightarrow q) \wedge (q \rightarrow P)$ |
|---|---|-------------------|-------------------|----------------------------------------------|
| T | T | T                 | T                 | T                                            |
| T | F | F                 | T                 | F                                            |
| F | T | T                 | F                 | F                                            |
| F | F | T                 | T                 | T                                            |

Create  
formula  
for  
bimplication  
using these  
truth  
combinations



$$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p) \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$$

$$p \rightarrow q \equiv \neg p \vee q$$

same

| p | q | $p \rightarrow q$ | $\sim p$ | $\sim p \vee q$ |
|---|---|-------------------|----------|-----------------|
| T | T | T                 | F        | T               |
| T | F | F                 | F        | F               |
| F | T | T                 | T        | T               |
| F | F | T                 | T        | T               |

$$p \rightarrow q$$

- $p$  implies  $q$
- if  $p$  then  $q$
- $q$  if  $p$  /  $q$  when  $p$
- $p$  only if  $q$
- $q$  follows from  $p$
- $q$  unless  $\neg p$

- $p$  is sufficient condition for  $q$ .
- $q$  is necessary condition for  $p$ .

$$p \leftrightarrow q$$

- $p \text{ iff } q$

- $p$  is necessary and sufficient condition for  $q$

Implication:  $p \rightarrow q$

Converse:  $q \rightarrow p$

Inverse:  $\neg p \rightarrow \neg q$

Contrapositive:  $\neg q \rightarrow \neg p$

Note:

$$\underbrace{p \rightarrow q}_{\text{Implication}} \equiv \neg p \vee q \equiv \underbrace{\neg q \rightarrow \neg p}_{\text{Contrapositive}}$$

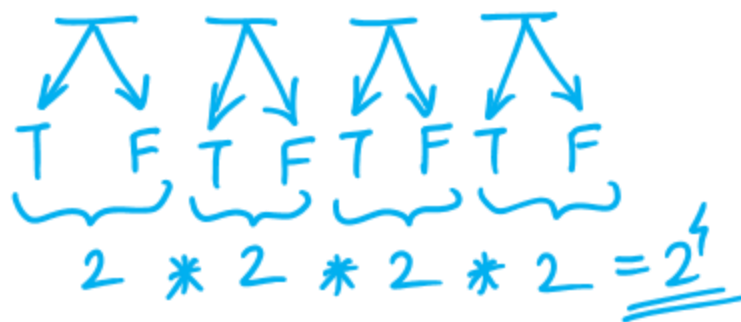
Q. How many propositional formulae are possible using 'n' propositional variables?



$4 = 2^2$

| P | Q |
|---|---|
| T | T |
| T | F |
| F | T |
| F | F |

For each row, there are 2 possible outcomes (T or F) for the formulae, indicated by red arrows and curly braces with the number 2.



$$2^4 = \underline{\underline{2^2}}$$

2 propositional variables  $\Rightarrow 2^2 = 2^4 = 16$  propositional formulae possible

3 propositional variables  $\Rightarrow 2^3 = 2^8 = 256$  propositional formulae possible

$\vdots$

$n$  propositional variables  $\Rightarrow \underline{\underline{2^{2^n}}}$  propositional formulae possible

$$p \rightarrow q \equiv \neg p \vee q \equiv \neg q \rightarrow \neg p$$

| $p$ | $q$ | $\neg p$ | $\neg q$ | $\neg p \vee q$ | $\neg q \rightarrow \neg p$ | $p \rightarrow q$ |
|-----|-----|----------|----------|-----------------|-----------------------------|-------------------|
| T   | T   | F        | F        | T               | T                           | T                 |
| T   | F   | F        | T        | F               | F                           | F                 |
| F   | T   | T        | F        | T               | T                           | T                 |
| F   | F   | T        | T        | T               | T                           | T                 |

same

Tautology: A proposition which is always true.

e.g.  $p \vee \neg p$

Contradiction: A proposition which is always false.

e.g.  $p \wedge \neg p$



Contingency: A proposition which is neither tautology nor contradiction.

e.g.  $P \wedge q$ ,  $P \vee q$ ,  $P \rightarrow q$

Satisfiability: A proposition which is true for atleast one truth combination.

e.g.  $P \wedge q$ ,  $P \vee q$ ,  $P \rightarrow q$

Q. What is the difference between Contingency & Satisfiability?

→ Contingency doesn't cover all true (tautology) case but Satisfiability covers the tautology case.



Dual: The dual of a compound proposition involving only the connectives  $\neg$ ,  $\wedge$  and  $\vee$  is obtained by replacing every

|          |      |          |
|----------|------|----------|
| $\wedge$ | with | $\vee$   |
| $\vee$   | with | $\wedge$ |
| T        | with | F        |
| F        | with | T        |

Note:  
Do not remove  
or add  $\neg$ .

$$A: p \wedge q \wedge r$$

$$A^* \text{ or } A^d: p \vee q \vee r$$

Note:

$$\underline{\underline{(A^d)^d \equiv A}}$$

$$A: \neg p \vee q \wedge r \vee \neg s$$

$$A^d: \neg p \wedge q \vee r \wedge \neg s$$

# Rules

## Idempotent

$$\begin{array}{l} p \vee p \equiv p \\ p \wedge p \equiv p \end{array}$$

dual

## Identity

$$\begin{array}{l} p \vee F \equiv p \\ p \wedge T \equiv p \end{array}$$

dual

## Domination

$$\begin{array}{l} \text{dual} \left\{ \begin{array}{l} p \vee T \equiv T \\ p \wedge F \equiv F \end{array} \right. \end{array}$$

## Commutative

$$\begin{array}{l} \text{dual} \left\{ \begin{array}{l} p \vee q \equiv q \vee p \\ p \wedge q \equiv q \wedge p \end{array} \right. \end{array}$$

## Negation

$$\begin{array}{l} \text{dual} \left\{ \begin{array}{l} p \vee \neg p \equiv T \\ p \wedge \neg p \equiv F \end{array} \right. \end{array}$$

## Associative

$$\begin{array}{l} \text{dual} \left\{ \begin{array}{l} p \vee (q \vee r) \equiv (p \vee q) \vee r \\ p \wedge (q \wedge r) \equiv (p \wedge q) \wedge r \end{array} \right. \end{array}$$

## De Morgan's

dual  $\begin{cases} \neg(p \vee q) \equiv \neg p \wedge \neg q \\ \neg(p \wedge q) \equiv \neg p \vee \neg q \end{cases}$

## Distributive

dual  $\begin{cases} p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r) \\ p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r) \end{cases}$

## Double Negation

$$\neg(\neg p) \equiv p$$

# Absorption

dual  $\begin{cases} p \vee (p \wedge q) \equiv p \\ p \wedge (p \vee q) \equiv p \end{cases}$

| $p$ | $q$ | $p \wedge q$ | $p \vee (p \wedge q)$ | $p \vee q$ | $p \wedge (p \vee q)$ |
|-----|-----|--------------|-----------------------|------------|-----------------------|
| T   | T   | T            | T                     | T          | T                     |
| T   | F   | F            | T                     | T          | T                     |
| F   | T   | F            | T                     | T          | F                     |
| F   | F   | F            | F                     | F          | F                     |



## Exploration Law

$$p \rightarrow (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$$

# Logical Equivalences involving Conditional Statements

1.  $p \rightarrow q \equiv \neg p \vee q$
2.  $p \rightarrow q \equiv \neg q \rightarrow \neg p$
3.  $p \vee q \equiv \neg p \rightarrow q$
4.  $p \wedge q \equiv \neg(p \rightarrow \neg q)$
5.  $\neg(p \rightarrow q) \equiv p \wedge \neg q$
6.  $(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$
7.  $(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$
8.  $(p \rightarrow q) \vee (p \rightarrow r) \equiv p \rightarrow (q \vee r)$
9.  $(p \rightarrow r) \vee (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$

## Example

What is the negation of the statement "Riya speaks Hindi and Riya speaks Telugu"?

The negation of an AND-statement is an OR of negations. Thus, the negation of this statement will be "Riya does not speak Hindi or Riya doesn't speak Telugu".

## Example

What is the negation of the statement "Riya speaks Hindi and Riya speaks Telugu"?

$$p \wedge q$$

The negation of an AND-statement is an OR of negations. Thus, the negation of this statement will be "Riya does not speak Hindi or Riya doesn't speak Telugu".

$p$ : Riya speaks Hindi.  
 $q$ : Riya speaks Telugu.

$$\neg(p \wedge q) \equiv \underline{\underline{\neg p \vee \neg q}}$$

## Example

What is an equivalent statement to “not (A and not B)”?

- not A and B
- A or not B
- A and not B
- not A or B

## Example

What is an equivalent statement to “not (A and not B)”?

- not A and B
- A or not B
- A and not B
- ✓ • not A or B

$$\neg (A \wedge \neg B) \equiv \neg A \vee B$$

## Example:

What's equivalent to “not (A and not (B or not C))”?

- not A or not B and not C
- not A or not B or not C
- not A or B or not C
- not A or B and not C

## Example:

What's equivalent to “not (A and not (B or not C))”?

- not A or not B and not C
- not A or not B or not C
- ✓ • not A or B or not C
- not A or B and not C

$$\neg(A \wedge \neg(B \vee \neg C))$$

$$\equiv \neg A \vee (B \vee \neg C)$$



# Predicate Logic (First Order Logic)

Proposition

$P$  : 2 is greater than 10. (F)

Predicate

$P(x)$  :  $x$  is greater than 10.  
Variable Predicate

$P(2) \Rightarrow (F)$

$P(12) \Rightarrow (T)$

## Predicate

When a proposition depends on variable(s), then it becomes a predicate.

## Domain

Whatever values the variables can take is generally called as domain.

## Predicate

→ Predicate is a proposition except for the fact that it contains variables whose values are to be taken from some universe of discourse.

$C(x): x \text{ is a cricketer.}$   
 $E(x): x + 2 \geq 5$

} 1-place predicate

$L(x, y): x \text{ likes } y.$   
 $E(x, y): x + y \leq 8$

} 2-place predicates

$f(x, y, z): x + y = z$

$T(p, q, t): p \text{ can teach } q \text{ concept at time } t.$

} 3-place predicates

→ A predicate involving 'n' variables is called  
n-place predicate.

$P(x): x > 3, x \text{ is real number.}$

Check for truth  $\left\{ \begin{array}{l} P(1) \\ P(2) \\ P(3) \\ P(\pi) \\ P(e) \end{array} \right.$

$P(x): x > 3$ ,  $x$  is real number.

Check for truth

- $P(1) \rightarrow \text{False}$
- $P(2) \rightarrow \text{False}$
- $P(3) \rightarrow \text{False}$
- $P(\pi) \rightarrow \text{True}$
- $P(e) \rightarrow \text{False}$

There are so many real numbers & deciding truth for each is difficult. We need to "quantify" the truth of the predicate  $P(x)$ .  
How much truth is there?

$\exists x P(x)$

$P(x): x^2 \geq 0$ ,  $x$  is real number.

Check for truth

- $P(1) \rightarrow \text{True}$
- $P(2) \rightarrow \text{True}$
- $P(3) \rightarrow \text{True}$
- $P(\pi) \rightarrow \text{True}$
- $P(e) \rightarrow \text{True}$

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How much truth is there?

?  $P(x)$



$P(x): x^2 \geq 0$ ,  $x$  is real number.

Check for truth

- $P(1) \rightarrow \text{True}$
- $P(2) \rightarrow \text{True}$
- $P(3) \rightarrow \text{True}$
- $P(\pi) \rightarrow \text{True}$
- $P(e) \rightarrow \text{True}$

There are so many real numbers & deciding truth for each is difficult. We need to "quantify" the truth of the predicate  $P(x)$ .  
How much truth is there?

$\forall x P(x)$

$Q(x, y): x + y = 1$  { This is a predicate. }

Is  $Q(1, 1)$  true?

$Q(1, 1): 1 + 1 = 1$  { Now, it is a proposition. }

Truth value: False

Is  $Q(-2, 1)$  true?

# Converting Predicates into Proposition

→ Using Substitution

# Quantifiers

→ Universal quantifier

$\forall$

for all

→ Existential quantifier

$\exists$

there exists

or

for some

e.g.  $U = \{1, 2\}$

$P(x) : x + 5 \leq 7$

For all  $x$ ,  $P(x)$  is true.

i.e.,  $\forall x P(x)$

Quantification of truth  
→ Always True

e.g.  $U = \{1, 2, 3\}$

$$P(x) : x + 5 \leq 7$$

Quantification {

- For some  $x$ ,  $P(x)$  is true.
- There exists  $x$  such that  $P(x)$  is true.
- There exists atleast one  $x$  such that  $P(x)$  is true.

i.e.,  $\exists x P(x)$   $\rightarrow$  Always True

Every student in this class studied Mathematics.

Case I:

$U$ : Set of students in this class.

$M(x)$ :  $x$  studied Mathematics.

$$\underline{\underline{\forall x M(x)}}$$

For all  $x$ ,  $M(x)$ .

## Case II:

$U$ : Set of all people.

$S(x)$ :  $x$  is student in this class.

$M(x)$ :  $x$  studied Mathematics.

$$\underline{\underline{\forall x [S(x) \rightarrow M(x)]}}$$

For all  $x$ ,  
if  $S(x)$  then  $M(x)$ .



# Negating Quantified Predicates

$$\textcircled{1} \neg \forall x P(x) \equiv \exists x \neg P(x)$$

$$\textcircled{2} \neg \exists x P(x) \equiv \forall x \neg P(x)$$

$$\textcircled{3} \overbrace{\forall x \neg P(x)}^{\curvearrowleft} \equiv \neg \exists x P(x)$$

$$\textcircled{4} \overbrace{\exists x \neg P(x)}^{\curvearrowleft} \equiv \neg \forall x P(x)$$

## Example 1: Rule (1)

$$\neg \forall x P(x) \equiv \exists x \neg P(x)$$

### Statement:

"All students passed the exam."  $\rightarrow \forall x P(x)$ , where  $P(x)$ : " $x$  passed the exam."

### Negation:

"Not all students passed the exam."  $\rightarrow \neg \forall x P(x)$ .

### Equivalent form:

"There exists at least one student who did not pass the exam."  $\rightarrow \exists x \neg P(x)$ .

## Example 2: Rule (2)

$$\neg \exists x P(x) \equiv \forall x \neg P(x)$$

### Statement:

"There is a student who scored full marks."  $\rightarrow \exists x P(x)$ , where  $P(x)$ : " $x$  scored full marks."

### Negation:

"There is no student who scored full marks."  $\rightarrow \neg \exists x P(x)$ .

### Equivalent form:

"Every student did not score full marks."  $\rightarrow \forall x \neg P(x)$ .

### Example 3: Rule (3)

$$\forall x \neg P(x) \equiv \neg \exists x P(x)$$

#### Statement:

"Every student failed the exam."  $\rightarrow \forall x \neg P(x)$ , where  $P(x)$ : " $x$  passed the exam."

#### Equivalent form:

"There does not exist a student who passed the exam."  $\rightarrow \neg \exists x P(x)$ .

## Example 4: Rule (4)

$$\exists x \neg P(x) \equiv \neg \forall x P(x)$$

### Statement:

"There exists a student who is not punctual."  $\rightarrow \exists x \neg P(x)$ , where  $P(x)$ : " $x$  is punctual."

### Equivalent form:

"Not all students are punctual."  $\rightarrow \neg \forall x P(x)$ .

# Aristotle Forms

All P's are Q's.

$$\forall x [P(x) \rightarrow Q(x)]$$

Note:  $\forall$  goes with  $\rightarrow$

Some P's are Q's.

$$\exists x [P(x) \wedge Q(x)]$$

Note:  $\exists$  goes with  $\wedge$

Not all P's are Q's.

$$\equiv \neg (\text{All P's are Q's})$$

$$\equiv \neg \{ \forall x [P(x) \rightarrow Q(x)] \}$$

$$\equiv \neg \{ \forall x [\neg P(x) \vee Q(x)] \}$$

$$\equiv \exists x [P(x) \wedge \neg Q(x)]$$

$$\equiv \text{Some P's are not Q's.}$$

No P's are Q's.

$$\equiv \neg (\text{Some P's are Q's})$$

$$\equiv \neg \{ \exists x [P(x) \wedge Q(x)] \}$$

$$\equiv \forall x [\neg P(x) \vee \neg Q(x)]$$

$$\equiv \forall x [P(x) \rightarrow \neg Q(x)]$$

$\equiv$  All P's are not Q's.



Some real numbers are rational.

$$\exists x [Re(x) \wedge Ra(x)]$$

Not all that glitters is gold.

$\equiv \neg (\text{All that glitters is gold})$

$\equiv \neg \{ \forall x [Gl(x) \rightarrow Go(x)] \}$

$\equiv \neg \{ \forall x [\neg Gl(x) \vee Go(x)] \}$

$\equiv \exists x [Gl(x) \wedge \neg Go(x)]$

$\equiv$  There exists a metal that glitters and is not gold.

None of my friends are perfect.

$\equiv \neg (\text{Some of my friends are perfect})$

$\equiv \neg \{ \exists x [F(x) \wedge P(x)] \}$

$\equiv \forall x [\neg F(x) \vee \neg P(x)]$

$\equiv \forall x [F(x) \rightarrow \neg P(x)]$

$\equiv \text{All of my friends are not perfect.}$

All purple mushrooms are poisonous. [Combined Sense]

$P_u(x)$ :  $x$  is purple.

$M(x)$ :  $x$  is mushroom.

$P_o(x)$ :  $x$  is poisonous.

For all  $x$ , if  $x$  is purple and  $x$  is mushroom,  
then  $x$  is poisonous.  $\forall x [(P_u(x) \wedge M(x)) \rightarrow P_o(x)]$

Gold and silver ornaments are precious. [Separate sense]

$G(x)$ :  $x$  is gold ornament.

$S(x)$ :  $x$  is silver ornament.

$P(x)$ :  $x$  is precious.

For all  $x$ , if  $x$  is gold or silver ornament, then  $x$  is precious.

$$\begin{aligned} & \forall x [(G(x) \rightarrow P(x)) \wedge (S(x) \rightarrow P(x))] \\ & \equiv \forall x [(G(x) \vee S(x)) \rightarrow P(x)] \end{aligned}$$

There is exactly one apple.

$$\exists x [A(x) \wedge \forall y [A(y) \rightarrow y = x]]$$

There exists  $x$  such that  $x$  is an apple & there exists  $y$  such that if  $y$  is an apple, then  $y = x$ .

There are exactly two apples.

$$\exists x \exists y [A(x) \wedge A(y) \wedge x \neq y \wedge \\ \forall z [A(z) \rightarrow z = x \vee z = y]]$$

There are at least two apples.

$$\exists x \exists y [A(x) \wedge A(y) \wedge x \neq y]$$



There is atmost one apple.

$$\text{Atmost one} \equiv \neg(\text{Atleast two})$$

$$\neg(\text{There are atleast two apples.})$$

$$\equiv \neg \exists x \exists y [A(x) \wedge A(y) \wedge x \neq y]$$

$$\equiv \forall x \forall y [\neg (A(x) \wedge A(y)) \vee x = y]$$

$$\equiv \forall x \forall y [(A(x) \wedge A(y)) \longrightarrow x = y]$$

# Provide feedback



# Thank You!